Rapidly Build and Deploy a Full-Stack App with Cursor

Advanced AI-assisted SaaS application development

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Cursor 3 & Context Management

AI-First Code Editor for Production Development

Not just autocomplete—full stack development

AI-First Code Editor for Production Development

Not just autocomplete—full stack development

⚡ Composer 2.5

Standard + Fast tiers—an iteration over Composer 2

🤖 Agents Window

Parallel agents across local, worktree, cloud & SSH

🐛 Bugbot

Automated PR review on Composer 2.5—~90-second reviews

🔗 MCP Integration

Connect to GitHub, Figma, databases

[1] [Cursor 3 Changelog]

Core Features

💬 Chat (Cmd+L)

- Multi-tab conversations
- Context checkpoints
- Quick Q&A

⌨️ Tab Autocomplete

- Multi-line predictions
- Context-aware
- Pattern learning

🎯 Agent (Cmd+I)

- Multi-file editing
- Autonomous execution
- Planning + building

✏️ Inline Edit (Cmd+K)

- In-place modifications
- Natural language
- Stay in flow

The Three-Step Framework

🔍

1. EXPLORE

Share context
Identify files
Discuss options

📋

2. PLAN

Step-by-step plan
Markdown checklists
Break down tasks

⚒️

3. BUILD

Execute iteratively
Review each step
Commit regularly

The Three-Step Framework

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💡 **This framework prevents context overload and ensures quality output**

[1] Cursor for Staff Engineers

Context Management: The Foundation

⚠️ Key Challenge

Context management is like telling a story—when it gets convoluted, the AI loses track

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⚠️ Key Challenge

Context management is like telling a story—when it gets convoluted, the AI loses track

Why It Matters

- Fixed context windows (128k-1M tokens)
- Comprehension ≠ token count
- Poor context = hallucinations

Best Practices

- Use @-mentions for precision
- Start new chats frequently
- Configure .cursorignore

The @ Symbol: Your Context Navigator

📄 @file

`@src/components/Header.tsx`

Reference specific files

📁 @folder

`@src/utils/`

Include entire directories

🔍 @codebase

`@codebase "authentication"`

Semantic search

📚 @docs

`@docs "React hooks"`

Query documentation

Effective Prompting

❌ Vague

```
Make this better
```

- No success criteria
- No direction
- Ambiguous goal

→

✅ Specific

```
@src/components/Button.tsx Refactor
to: 1. Add TypeScript props 2.
Include disabled state 3. Add
loading spinner 4. Keyboard
accessible
```

💡 **Describe the diff you want: current state → desired state**

Project Rules: .cursor/rules

Four Application Modes

1. **Always Apply** - Every session automatically
2. **Apply Intelligently** - AI determines relevance (recommended)
3. **Apply to Specific Files** - Pattern matching (*.test.ts)
4. **Apply Manually** - Activated with @-mention

Project Rules: .cursor/rules

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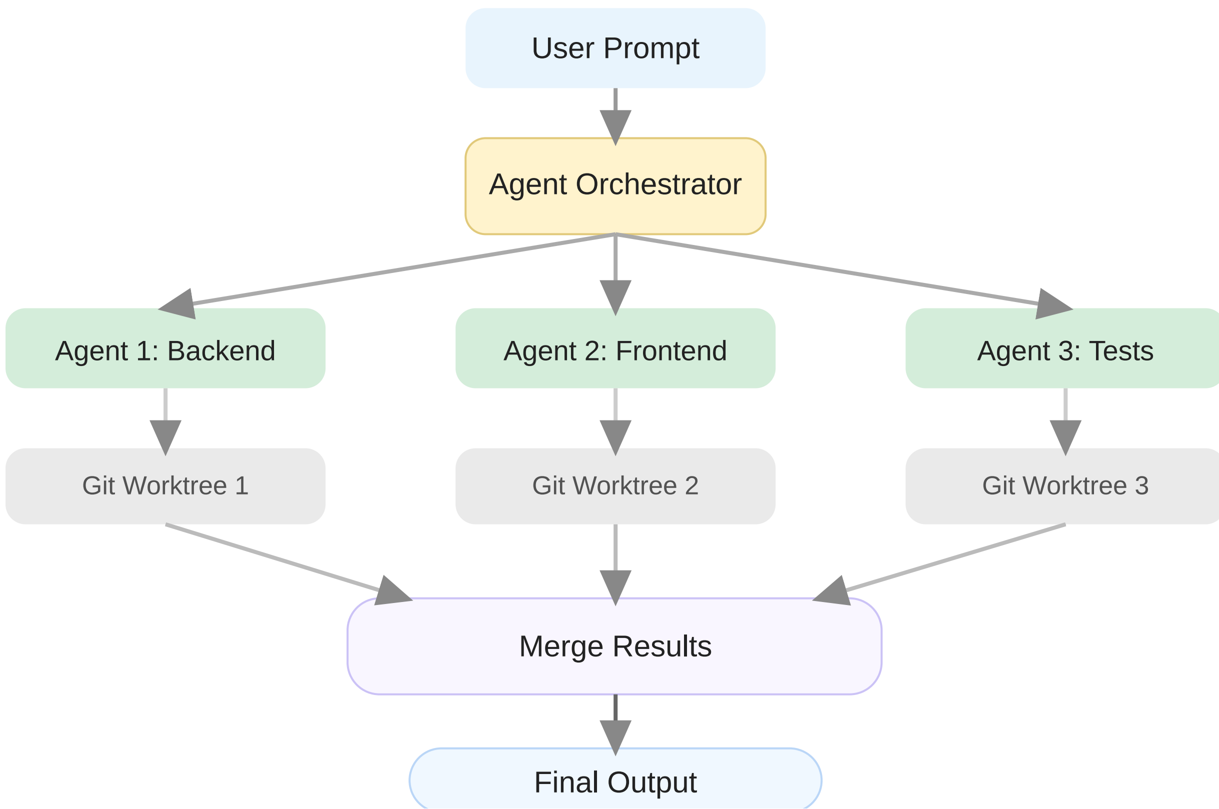
✅ When to Use

- Recurring mistakes
- Critical security patterns
- Team consistency

⚠️ Watch Out

- Can slow you down
- Keep under 500 lines
- Better context > more rules

Multi-Agent Architecture



Each agent operates in isolated git worktree → No conflicts

Model Context Protocol (MCP)

Open protocol connecting AI to tools & data

Like USB for AI—standardized interface

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Tools

Functions AI executes

Prompts

Templated workflows

Resources

Structured data sources

🐙 GitHub MCP

- Create PRs
- Review code
- Manage issues

```
npx @modelcontextprotocol/server-github
```

🎨 Figma MCP

- Extract design tokens
- Generate components
- Sync UI updates

```
bunx cursor-talk-to-figma-mcp
```

🔗 GitMCP

- Turn any repo into MCP
- Replace github.com → gitmcp.io
- Instant context access

```
gitmcp.io/facebook/react
```

🗄️ Database MCPs

- PostgreSQL
- SQLite
- Query and inspect

```
@modelcontextprotocol/server-postgres
```

[1]GitMCP | [2]Figma MCP

Demo: Cursor Configuration Setup

📄 See: `presentation/scripts.md`

Demo 1: Setting Up Cursor Configuration

Questions?

Open floor—ask about anything from this section

Break

10 Minutes ☕

Building Your First Full App with Cursor

AI-Driven Project Initialization

📝

1. Describe

Problem, users, features in plain English

→

🛠️

2. Define

Tech stack, database, auth, deployment

→

🚀

3. Generate

Let Cursor scaffold everything

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The Context File Approach

- Create `docs/context.md` with your vision
- Reference it inside prompts: `'Generate the project structure for @docs/context.md '`

[1]Building Apps with AI - Complete Workflow

Example: Context File

```
# DeepWork AI - Focus Productivity App

## Problem
Developers struggle with multitasking and distractions.
Need a tool to prioritize tasks and maintain deep work focus.

## Users
Software engineers, designers, product managers

## Core Features
1. Task management with priority levels
2. Focus timer for single-task concentration
3. AI chat to quickly add tasks via natural language

## Tech Stack
- Frontend: Next.js 15+ (App Router), TypeScript, Tailwind
- Backend: Supabase (database + auth)
- AI: OpenAI API for chat
- Deployment: Vercel

## Database Schema
[Let Cursor generate based on features]
```

ChatGPT + Cursor Workflow

1️⃣ ChatGPT

Planning Phase:

- Brainstorm features
- Structure ideas
- Create context.md
- Plan architecture

→

2️⃣ Cursor

Implementation:

- Generate code
- Multi-file operations
- Debug errors
- Refactor

💡 **Use ChatGPT (or Claude/Gemini/etc.) for exploration, Cursor for implementation**

Demo: Building Your First Full App

Setting Up Supabase Backend

1. Create Project

- Visit supabase.com
- Create new project
- Copy API URL & anon key

2. Environment Variables

```
NEXT_PUBLIC_SUPABASE_URL=...
NEXT_PUBLIC_SUPABASE_ANON_KEY=...
```

3. Prompt Cursor

```
@docs/context.md

Install Supabase client and create
the database schema for tasks with:
title, description, priority,
deadline, completed status, user_id.
```

✅ **Cursor generates SQL migrations + RLS policies**

Creating UI Components

Prompt for Component Generation:

```
Create the following components following our project rules:

1. TaskCard - Display single task with priority badge
2. TaskList - Grid of TaskCards with filtering
3. AddTaskModal - Form to create new tasks
4. FocusTimer - Countdown timer for focused work

Use Tailwind, make them responsive, and include
TypeScript interfaces for all props.
```

✅ Agent Mode Delivers:

💡 Pro Tip:

Iterative Development Loop

1. Generate code with Agent or Composer

↓

2. Run the app - Check for errors: `npm run dev`

↓

3. Copy error messages - Paste into Cursor chat

↓

4. Let Cursor fix - Review the changes

↓

5. Commit - Save working state with git

[1] Complete App Building Workflow

Testing with AI Assistance

Generate Test Suites:

```
@src/components/TaskCard.tsx
  Create a complete test suite using Jest and React Testing Library.
  Include:
  - Rendering tests
  - User interaction tests
  - Edge cases
  - Accessibility tests
```

Testing with AI Assistance

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🧪

Component Tests

🔗

Integration Tests

🌐

E2E Tests

Questions?

Open floor—ask about anything from this section

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Advanced Architecture Patterns

🏗️ Atomic Design

- **Atoms** - Button, Input
- **Molecules** - SearchBar
- **Organisms** - Header
- **Templates** - Layouts
- **Pages** - Full views

🗂️ State Management

- **Local** - useState
- **Global Client** - Zustand
- **Server** - React Query
- **URL** - searchParams

🎣 Custom Hooks

- Single responsibility
- Named exports
- TypeScript interfaces
- Reusable logic

⚡ Performance

- React.memo()
- useMemo()
- useCallback()
- Code splitting

Demo: AI-Powered Quiz App

Quiz App: Tech Stack

AI-Generated Quiz Application

Frontend

Next.js 15+
TypeScript
Tailwind CSS

AI

OpenAI API
Instructor
Structured Output

Backend

Supabase
Real-time
Row Level Security

Architecture Diagrams with Mermaid

Prompt Cursor to Generate Diagrams:

```
Create a Mermaid diagram showing the data flow in our quiz app:<br>
- User selects topic<br>
- API generates questions with OpenAI<br>
- Questions displayed one at a time<br>
- Answers stored in Supabase<br>
- Results calculated and shown
```

Architecture Diagrams with Mermaid

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✅ Diagram Types

- flowchart - Logic flow
- sequenceDiagram - Interactions
- classDiagram - Data structures

💡 Best Practice

Start small, layer upward, multiple perspectives

[1] [Cursor Cookbook - Mermaid Diagrams]

Figma MCP Integration

Figma Design

- Design tokens
- Component specs
- Typography

→

React Code

- Pixel-perfect
- Tailwind classes
- Responsive

Prompt Example:

```
@figma file:quiz-app-designs

Generate the QuestionCard component matching the Figma design.
Extract all design tokens (colors, spacing, shadows) and
create a matching React component with Tailwind.
```

[1]Cursor Talk to Figma MCP

Questions?

Open floor—ask about anything from this section

Full-Stack Application

Backend API Development

Prompt for Complete REST API:

```
Create a complete REST API for our quiz app:

Endpoints:
- POST /api/quiz/generate - Create new quiz
- GET /api/quiz/[id] - Get quiz by ID
- POST /api/quiz/[id]/answer - Submit answer
- GET /api/quiz/[id]/results - Get final results
- GET /api/quiz/history - User's past quizzes

Include: Authentication, validation (Zod), error handling,
rate limiting, API documentation comments
```

Backend API Development

🔐

Auth

✅

Validation

⚡

Rate Limiting

Database Optimization

Optimization Strategies

- **Indexes** - Speed up queries
- **JOINs** - Reduce round trips
- **Pagination** - Limit results
- **Caching** - Redis layer
- **Connection Pooling** - Supavisor

Prompt Cursor:

```
@src/lib/db/queries.ts

Optimize this query for performance:
- Add proper indexes
- Use JOIN instead of multiple queries
- Implement pagination
- Add caching strategy
```

State Management & Real-time

📊

Zustand Store

Quiz state, answers, timer

⚡

Optimistic Updates

Instant UI feedback

🔄

Real-time Sync

Supabase subscriptions

Optimistic Update Pattern

1. Update UI immediately (optimistic)
2. Send request to server
3. If error: rollback UI changes
4. If success: keep UI as-is

Result: App feels instant ⚡

Demo: Test, Iterate, Deploy

Questions?

Open floor—ask about anything from this section

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10 Minutes ☕

Deployment & Production

Deployment with Vercel

From Code to Production in 10 Minutes

🚀 Zero Config

Auto optimizations

🌍 Edge Network

Global CDN

🔄 Instant Rollback

One-click revert

CI/CD Pipeline

Generate GitHub Actions Workflow:

```
Create a GitHub Actions workflow for our Next.js app:

On every PR:
1. TypeScript type checking
2. ESLint
3. Tests (unit + integration)
4. Build
5. Deploy preview to Vercel

On merge to main:
All above + E2E tests + Production deploy + Slack notification
```

CI/CD Pipeline

✅ **Type Check**

🧪 **Test**

🚀 **Deploy**

Production Readiness Checklist

🔐 Security

- Env vars secured
- SSL + CORS configured
- Rate limiting enabled
- Dependency & secret scanning

⚡ Performance

- CDN + edge caching
- Image optimization
- Code splitting
- Database indexes

✅ Code Quality

- Type safety (strict TS)
- ESLint + a11y checks
- 80% test coverage

📊 Monitoring & Recovery

- Error tracking (Sentry)
- Uptime + analytics
- Automated backups
- Rollback strategy

Automated PR Workflow

Staff Engineer Tip: Handle PR Feedback Fast

📸

1. Screenshot

Capture PR comments

→

🤖

2. Cursor Plan

Drag into Cursor, generate plan

→

⚒️

3. Execute

Fix items one by one

Even faster: run `/review` (`/review-bugbot`, `/review-security`) before you push—Bugbot integrates with GitHub & GitLab PRs

[1] Automating PR Feedback Workflow | [2] Cursor Changelog—/review

Demo: Vercel & Github Actions

Cost Optimization Strategies

💸 Current Costs

- Vercel Pro: $20/month
- Supabase Pro: $25/month
- OpenAI API: ~$50/month
- Upstash Redis: $10/month

Total: ~$105/month

💰 Optimizations

- ✅ Switch to DeepSeek (10x cheaper)
- ✅ Aggressive caching
- ✅ Free tier for previews
- ✅ Optimize DB queries
- ✅ Rate limiting

New Total: ~$40/month

💡 Prompt Cursor for Cost Analysis

Analyze our app's costs and suggest optimizations to reduce spending while maintaining performance.

Key Takeaways

✅ Context is King

Strategic @-mentions, clean project structure

✅ Three-Step Framework

Explore → Plan → Build

✅ Production-Ready

Testing, CI/CD, monitoring, security

✅ MCP Integration

GitHub, Figma, databases

🎯 Remember

Cursor augments your workflow—it doesn't replace engineering judgment. Use it to eliminate tedious work and focus on what matters.

Additional Resources

📚 Documentation

- Cursor Docs
- Changelog
- Composer 2.5

🔗 MCP Servers

- Official Servers
- GitMCP
- Figma MCP

🎥 Videos

- Staff Engineers
- Building Apps

Connect With Me

📚 Course materials

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